Unit Assessment 2

Newton's Laws

Instructions

- **There are 4 problems. Do at least 3.** You may do all 4 for extra credit. If you do all 4, your three highest-scoring problems count fully, and you earn half of your score on the lowest-scoring problem as extra credit (up to 16.5 pts).
- **The test is out of 100 pts. Each problem is worth 33 pts**, and you get 1 pt for writing your name.
- **Show your work neatly and in an easy-to-follow way.** (This helps me give you points!)
- **Clearly circle or box your answer.** For multipart problems, circle or box the answer to each part.
- **If you run out of space, use the back or another sheet of paper.** Note in your work where you continued it, so I can find it and give you all the points you deserve!

Full Name: _____

Coyote ID: _____

Equations

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \qquad \cos\theta = \frac{\text{adj}}{\text{hyp}} \qquad \tan\theta = \frac{\text{opp}}{\text{adj}}$$

$$\vec{F}_{\text{net}} = m_{\text{total}}\vec{a} \qquad |\vec{F}_G| = |\vec{w}| = mg \qquad g = 9.8\,\text{m/s}^2$$

$$v_t = \omega R \qquad a_t = \alpha R \qquad a_r = \frac{v_t^2}{R} = \omega^2 R$$

$$v_f = v_i + a\Delta t \qquad s_f = s_i + v_{i,s}\Delta t + \frac{1}{2}a_s\Delta t^2 \qquad v_f^2 = v_i^2 + 2a\Delta s$$

$$f_{s,\max} = \mu_s n \qquad f_k = \mu_k n$$